

Fundamentals of functional and effective connectivity and their applications to psychotic disorders

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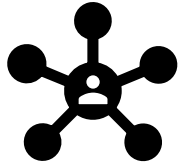


Institute
for Mental
Health

A note about this presentation



Talking about or listening to certain topics can be difficult to some people. It can raise emotions and memories, and everyone deals differently with this.



This is a safe and respectful space. If you need some physical or psychological distance, you are welcome to leave the room and join again at any point during the presentation.

Outline

Relevance of
cognitive function
and functional large-
scale networks

Theoretical basis of
functional
connectivity with
examples

Theoretical basis of
effective
connectivity with
examples

Advantages and
disadvantages of
both approaches

Discussion

Summary

Learning objectives

By the end of this session, you will be able to:

Understand the difference between functional and effective connectivity

Consider the advantages and disadvantages of functional and effective approaches in mental health

Critically appraise the potentials and limits of both approaches in mental health



- Are functioning brain networks important?
- Should we be worried if our brain networks are not working optimally? How do we know if they may not work optimally?
- Should we be worried if our brain networks are not working optimally when we have a mental health condition?

Relevance of cognitive function and functional large-scale networks in psychotic disorders

Types of Psychotic Disorders

1

SCHIZOPHRENIA:

A chronic mental illness causing hallucinations, delusions, and disorganized thinking, often linked to genetics and brain structure.

2

SCHIZOAFFECTIVE DISORDER:

Combines psychotic symptoms with mood disturbances like depression or mania.

3

SCHIZOPHRENIFORM DISORDER:

Short-term schizophrenia-like symptoms lasting less than six months.

4

BRIEF PSYCHOTIC DISORDER:

A sudden, short episode of psychosis triggered by stress, resolving within a month.

5

SUBSTANCE-INDUCED PSYCHOTIC DISORDER:

Psychosis triggered by drugs or withdrawal, which subsides after detox.

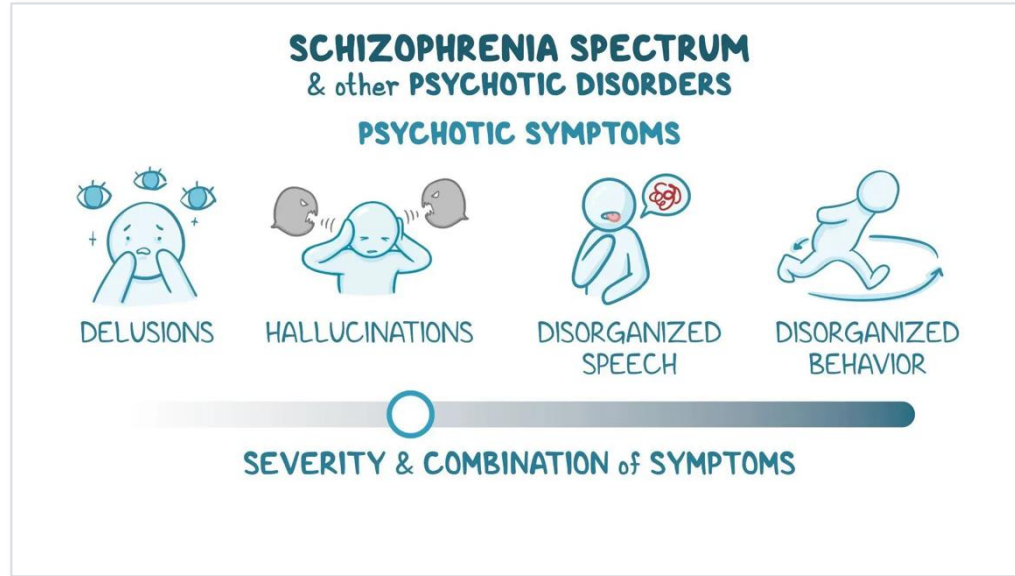
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DELUSIONAL DISORDER:

Persistent false beliefs with otherwise normal behavior and functioning.

Psychotic disorders - Background

- Prevalence and diagnosis
 - Approximately 0.5 - 1% or approximately 24 million people worldwide
- Neurodevelopmental and neurobiological disorder
 - Onset typically during late adolescence or early adulthood
 - Onset tends to happen earlier among men than women



Psychotic disorders - Background



- Symptoms and diagnosis
 - Positive symptoms (i.e., hallucination, delusions)
 - Negative symptoms (i.e., alogia)
 - Cognitive deficits (i.e., verbal memory, working memory, emotion recognition)
- Some individuals are at high risk and about 22% of these individuals will transition to a psychotic disorder.
- Already prior to diagnosis, cognitive deficits can occur, which are also prevalent with illness progression and are associated with social disability.



Psychotic disorders - Multifactorial Causes



Genetic factors

- Genetic risk
- Familial risk

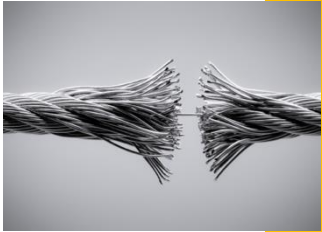
Neurobiological factors

- Structural, functional and network alterations
- Microbiological changes



Environmental factors

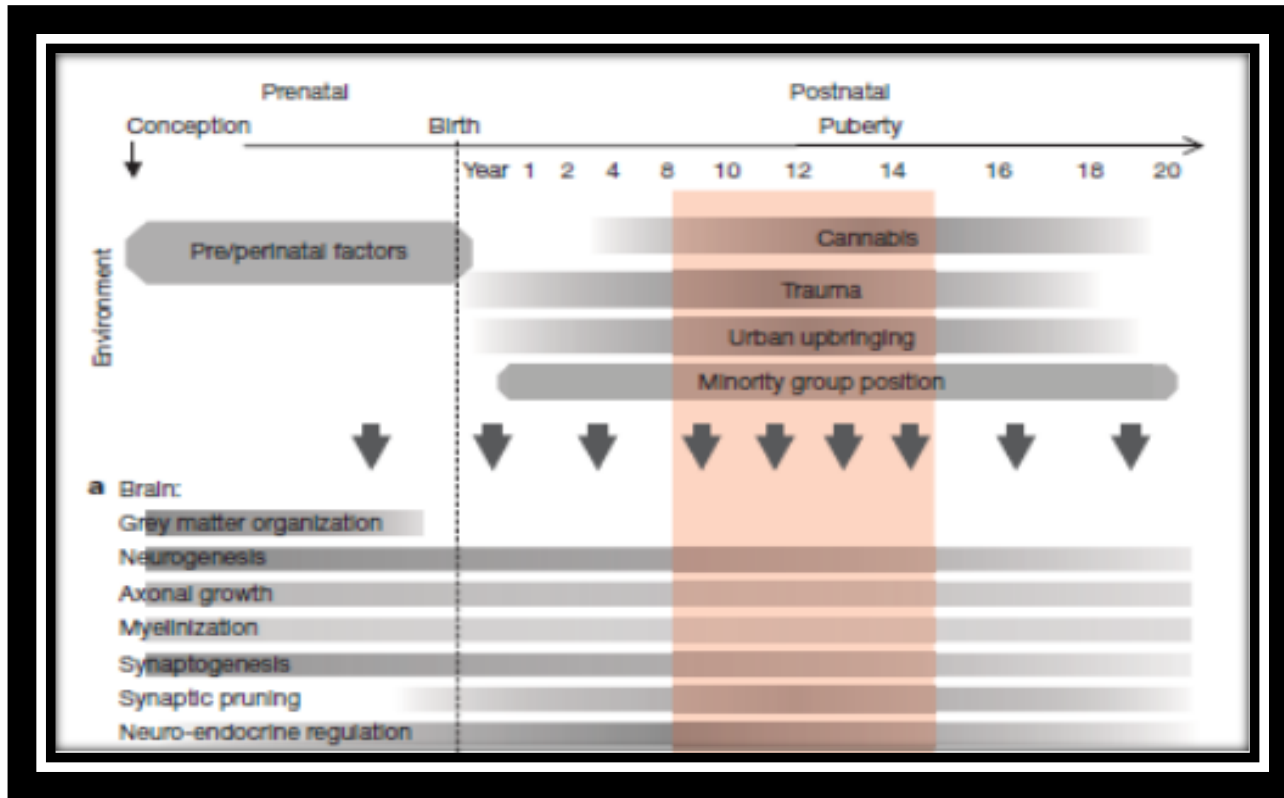
- Childhood trauma
- Stress
- Infections
- Migrations



Other factors

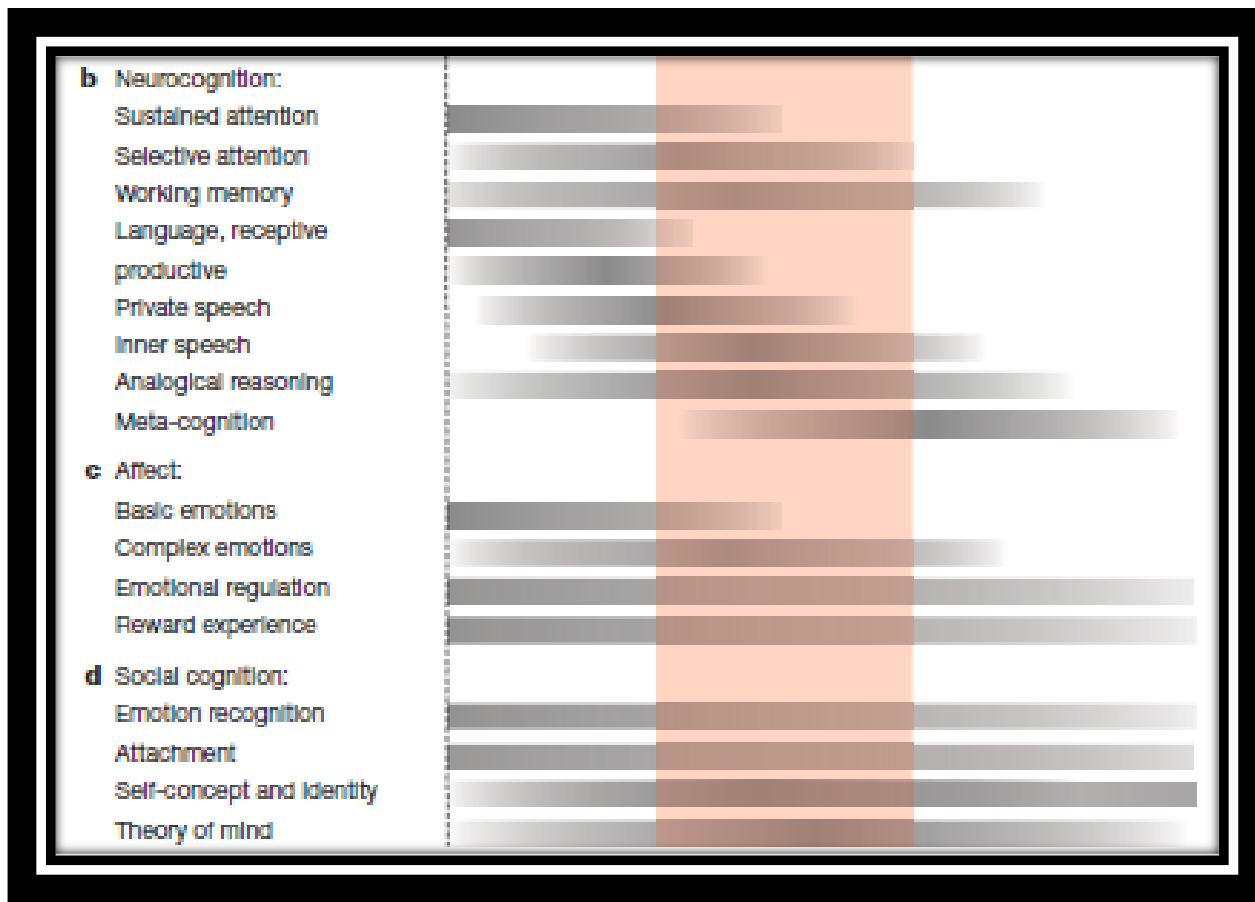
- Substance abuse
- Sleep deprivation
- Childbirth complications





Environmental
and biological risk
factors in
psychosis

Approximate timing of the development of the
human brain, functional abilities, and impact of
environmental exposures



Impact on
psychological
and cognitive
factors

Approximate timing of the development of the human brain,
functional abilities, and impact of environmental exposures

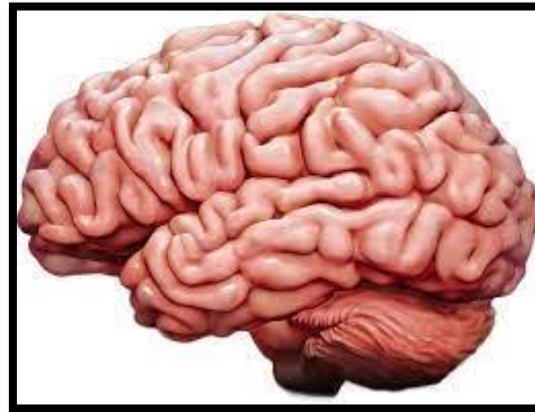
Characteristics of cognitive function in individuals in psychotic disorders

Clinical and cognitive symptoms



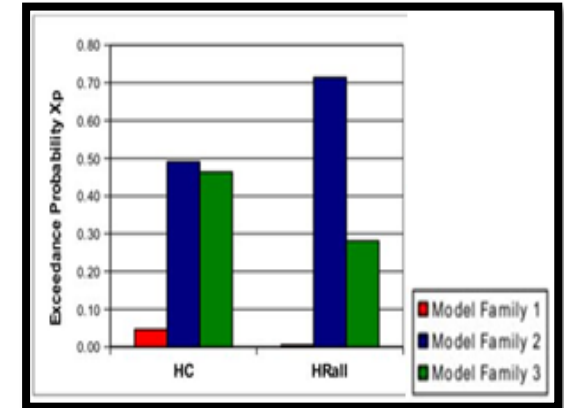
- Clinical symptoms
- Cognitive deficits

Neurobiological markers



- Lower surface area
- Lower cortical thickness
- Smaller volumes

Functional networks



- Functional and effective connectivity



Theoretical basis of functional connectivity with examples

Functional Segregation

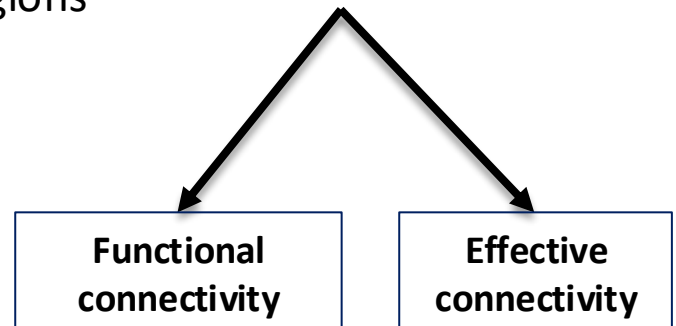
Specialised areas exist in the cortex

- Analyses of regionally specific effects
- Identifies regions specialised for a particular task
- Univariate analysis

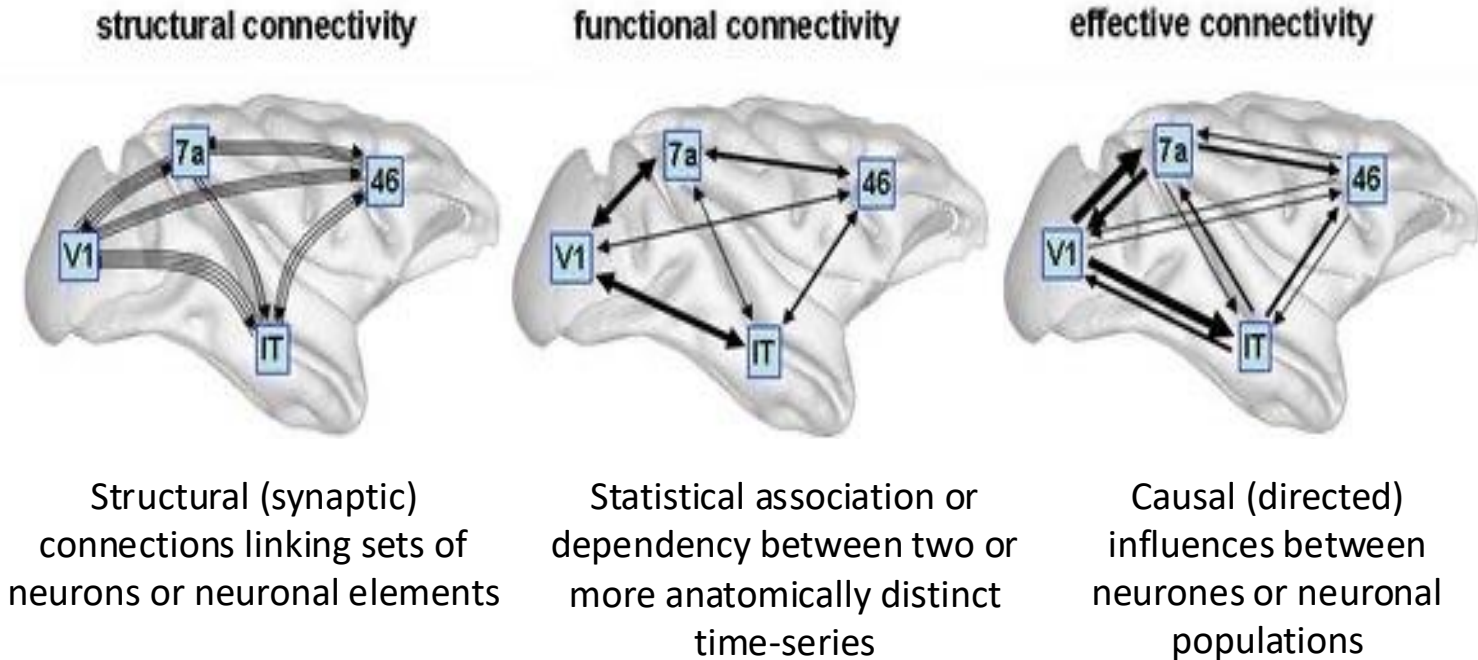
Functional Integration

Network(s) of interactions among specialised areas

- Analysis of how different regions in a neuronal system interact
- Determines how experimental manipulation affects coupling between regions

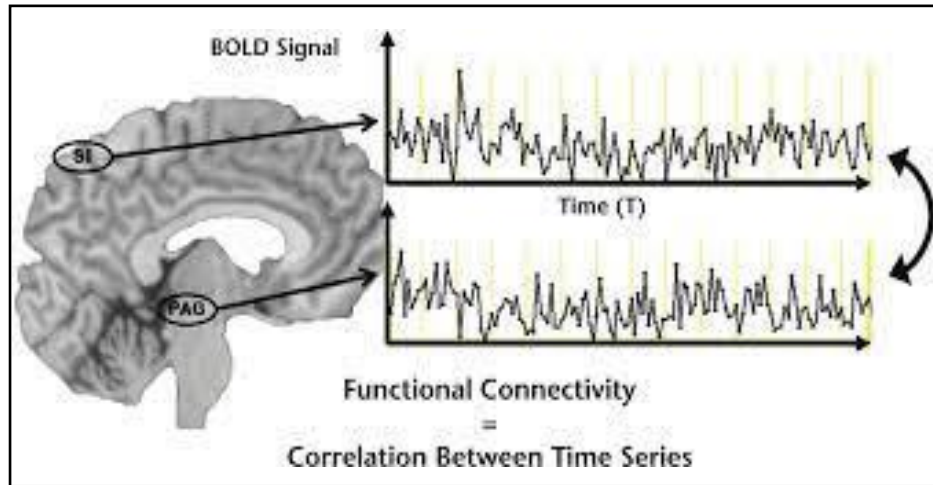


Functional neuroimaging with computational modelling



Functional connectivity

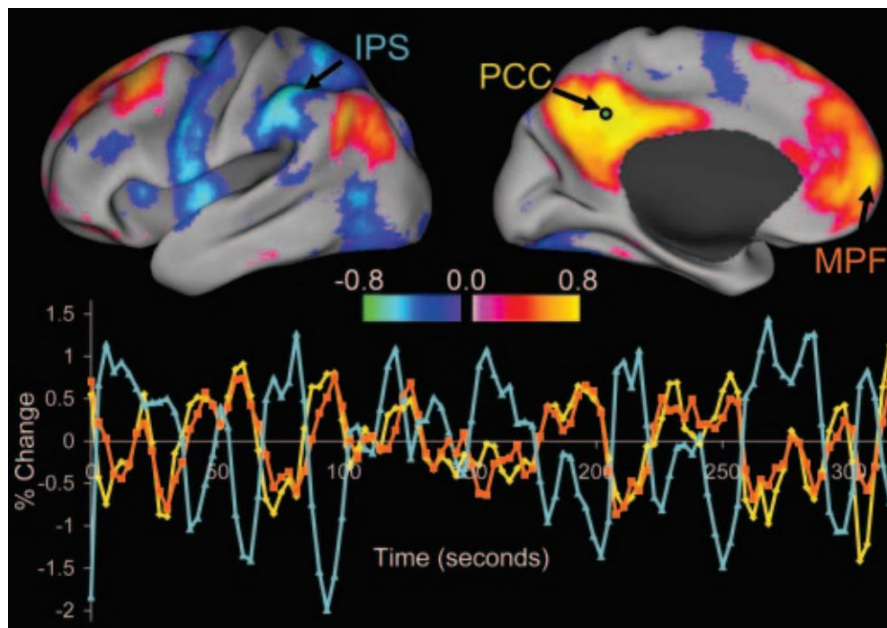
Definition: Statistical associations or dependencies between regional time series



Extraction of reference of time series

Voxel-wise correlation with time series from other voxels in the brain

Functional connectivity

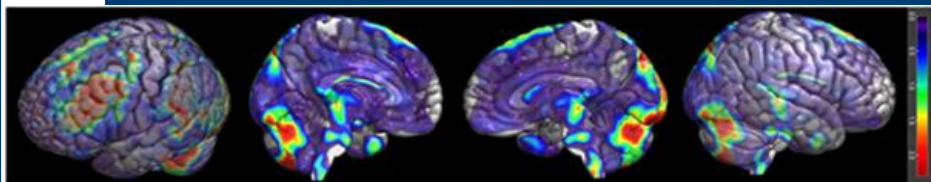


- Occurs during task and at rest
- Resting-state networks
 - Correlation between spontaneous BOLD signals of brain regions known to be functionally and/or structurally related
- Default mode network (DMN)
 - Decreased activity during cognitive tasks
 - Inversely related to regions activated by cognitive tasks
- Task-positive and task-negative networks

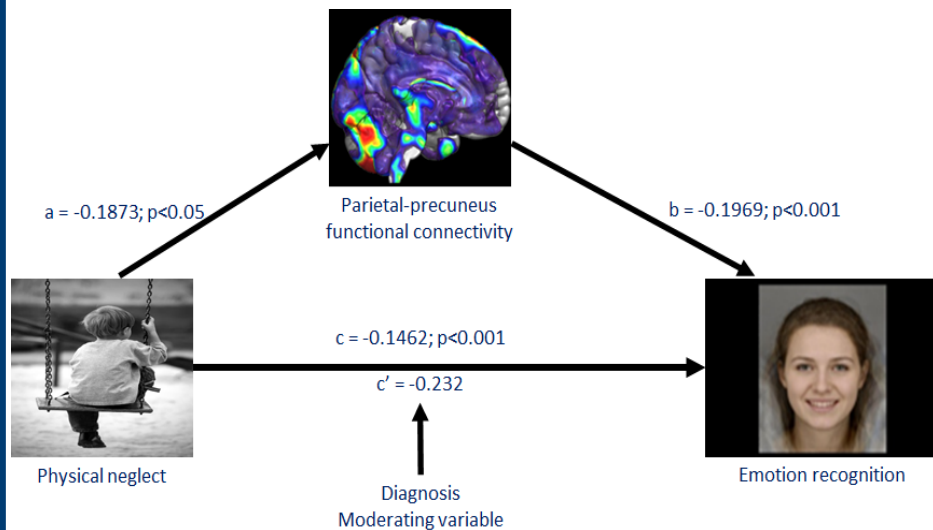
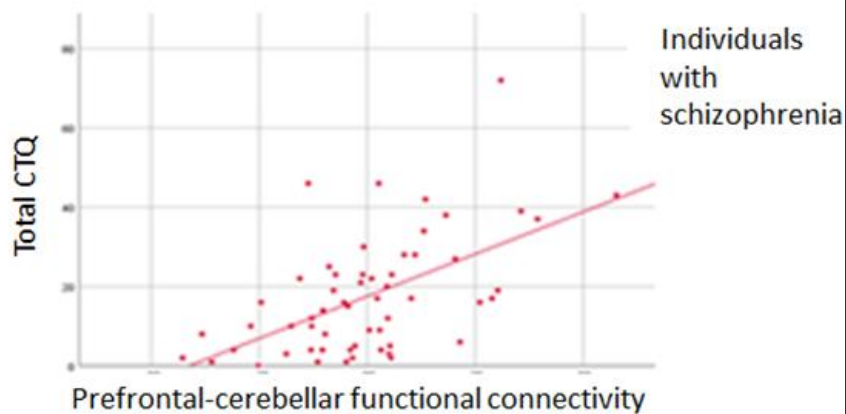
Functional connectivity approaches

- Model-dependent method - *A priori* or hypothesis-driven from previous literature
- Model-free method – Data-driven method
- Seed-based functional connectivity analysis
- Coherence analysis (EEG)
- Eigen-composition (i.e., Principal Component Analysis)
- Independent Component Analysis

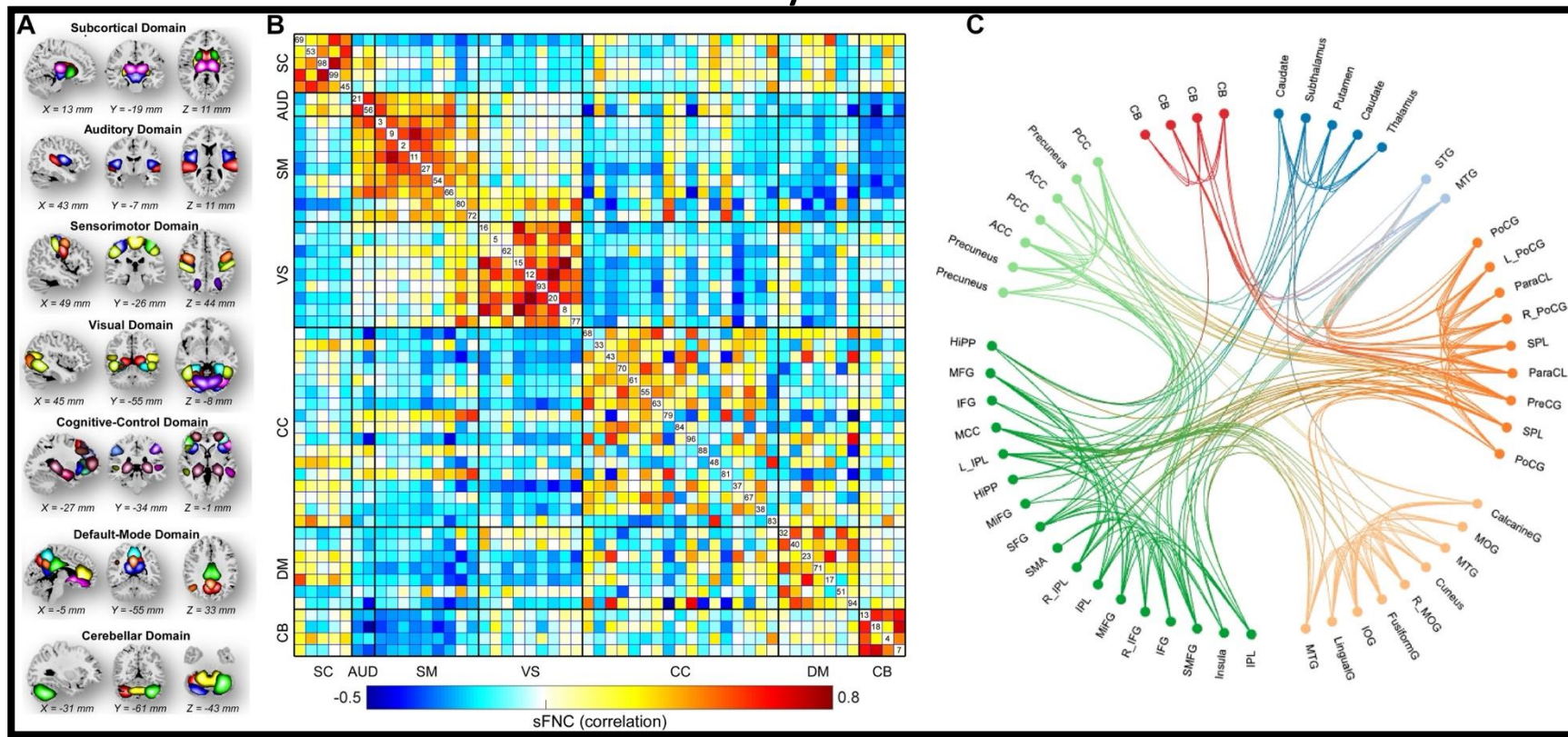
CHILDHOOD ADVERSITY AND RESTING-STATE FUNCTIONAL CONNECTIVITY



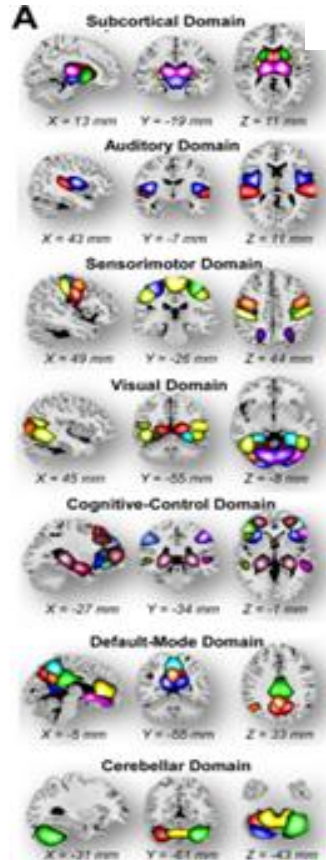
Right lateral parietal lobe seed



Functional connectivity - EEG



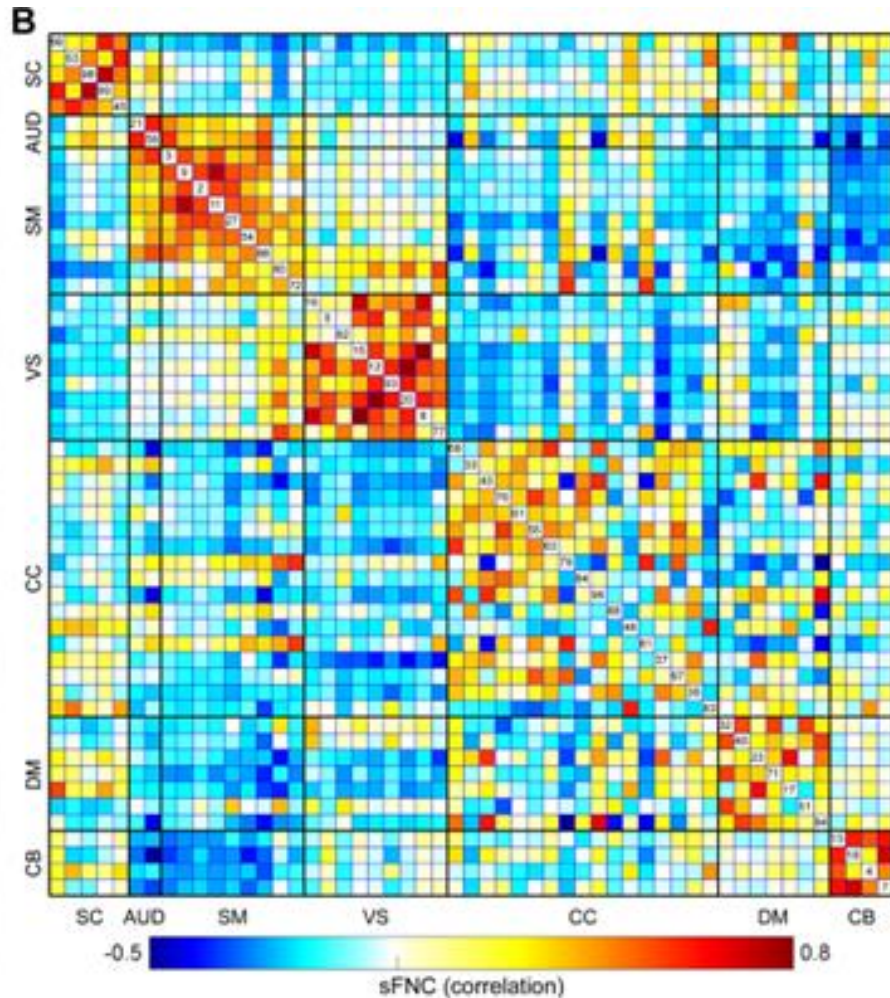
Spatial maps of the identified intrinsic connectivity networks and spatial functional network connectivity



53 ICNs were identified and sorted into seven resting-state functional domains

Large-scale group of distinct brain regions that show synchronised, spontaneous activity over time

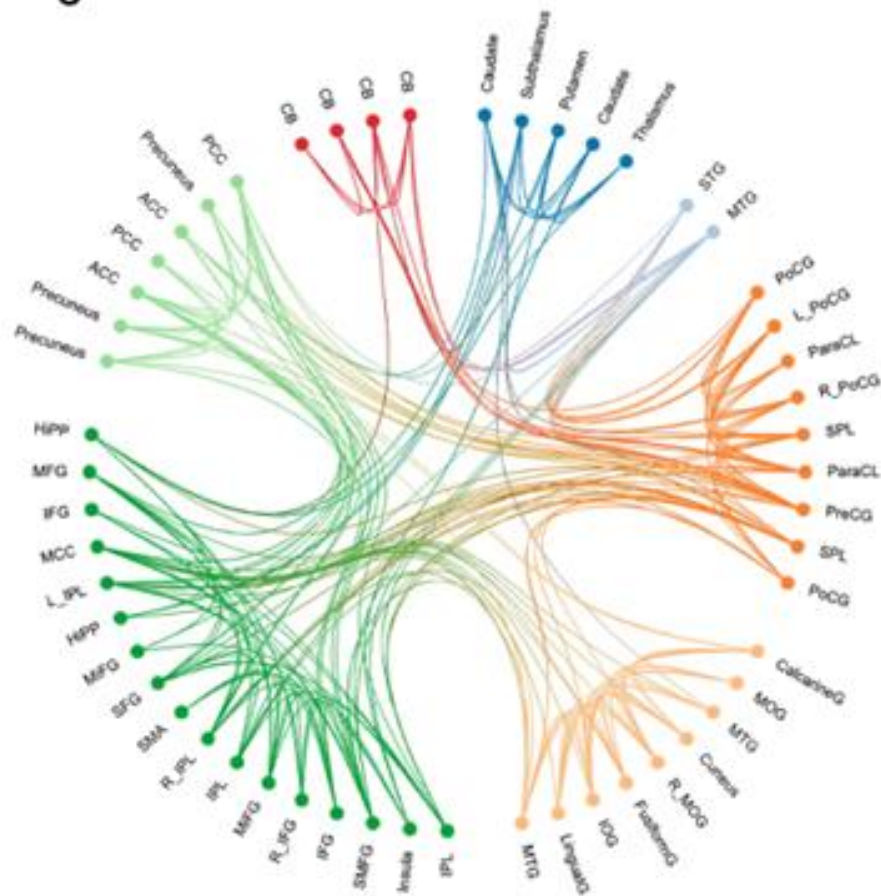
Each colour represents a single intrinsic connectivity network



Averaged spatial functional network connectivity matrix across participants

Measurement of higher-order statistical relationships and dynamic changes between distinct spatial brain components or networks over time

C



The functional connectivity profile of the averaged spatial functional network connectivity matrix

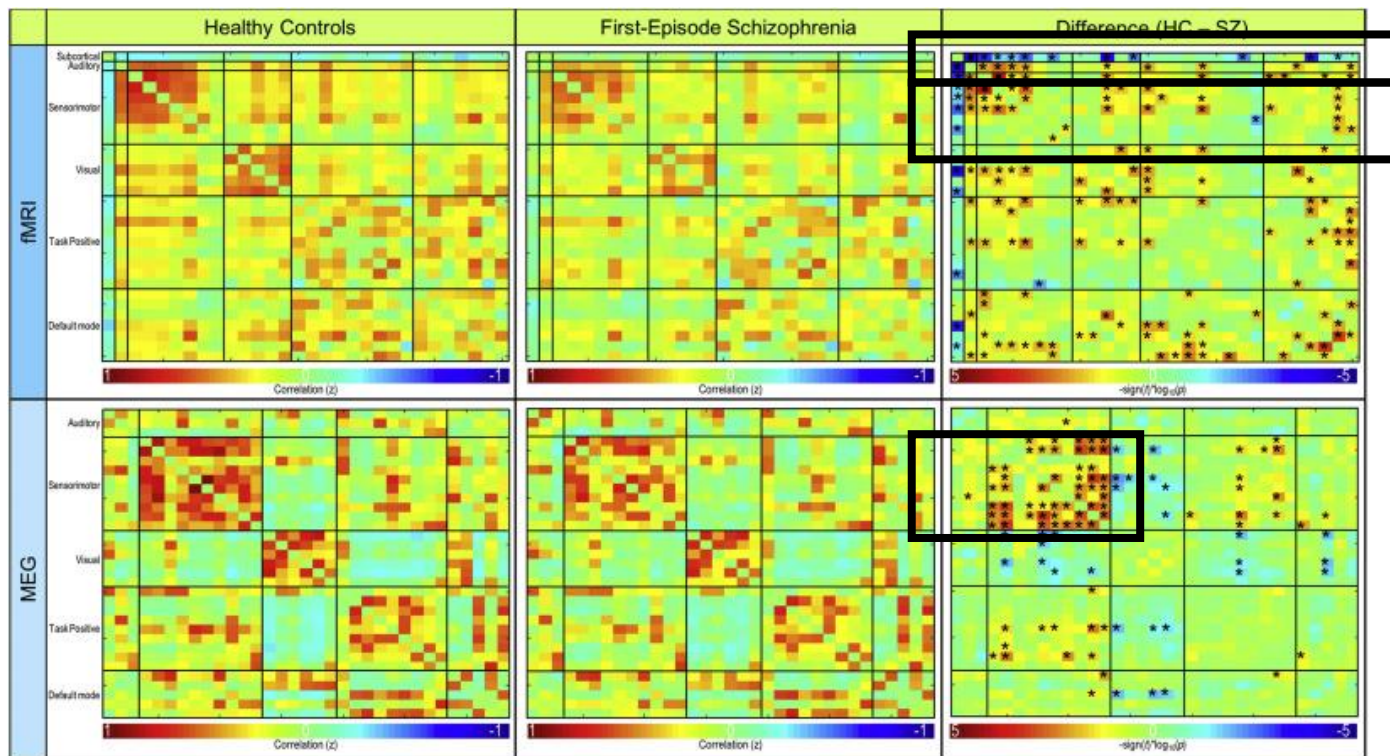
Indicating more unstable brain connections and several networks affected at the same time

Altered neural oscillations and synchrony in schizophrenia - MEG

Neural Oscillation	Anatomy	Function
Delta (1-3 Hz) 	Thalamus, Neocortex	Memory, Synaptic plasticity
Theta (4-7 Hz) 	Hippocampus, Sensory cortex, Prefrontal cortex	Memory, Synaptic plasticity, Top-down control, Long-range synchronization
Alpha (8-12 Hz) 	Thalamus, Hippocampus, Reticular formation, Sensory cortex, Motor cortex	Inhibition, Attention, Consciousness, Top-down control, Long-range synchronization
Beta (13-30 Hz) 	All cortical structures, Subthalamic nucleus, Basal ganglia, Olfactory bulb	Sensory gating, Attention, Motor control Long-range synchronization
Gamma (30-200 Hz) 	All cortical structures, Basal ganglia, Retina, Olfactory bulb	Perception, Attention, Memory, Consciousness Synaptic plasticity

- Changes to neural oscillations and their synchronisation may be core pathophysiological mechanisms in schizophrenia
- Synchronisation of neural oscillations is the primary way to measure functional connectivity
- When two distant brain regions synchronise their rhythmic wave patterns (oscillations), they show functional connectivity.

Resting-state fMRI (7T) and MEG Functional Connectivity



Importance of examining multimodal neuroimaging data as critical disorder-related information may not be detectable in a single modality alone

FDR-corrected significant differences ($p_{FDR} < 0.05$) are indicated with •
and uncorrected significant differences ($p_{uncorrected} < 0.05$) are indicated with *.



Theoretical basis of effective connectivity with
examples

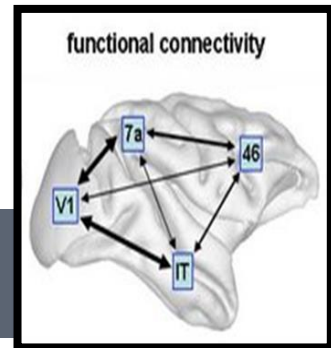
Functional connectivity

Advantages

- No experimental control (i.e., resting-state but also useful for tasks)
 - Free from experimental confounds
 - Possible to scan participants who would find it difficult to complete a task
 - No model of what caused the data (i.e., sleep or hallucinations)
 - Testing the 'baseline' hypothesis prior to experimental task
 - Complex networks

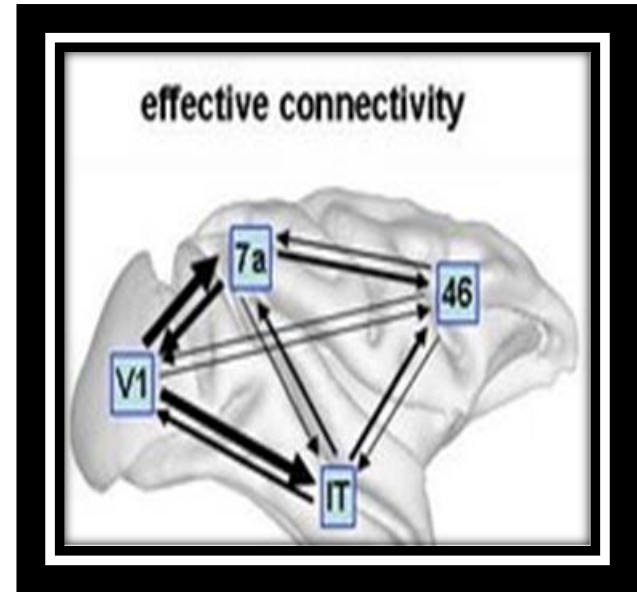
Disadvantages

- Interpretation of findings is based on correlations; descriptive
- No mechanistic insight
- Typically, dissatisfying for experimental tasks
- Typically, suboptimal for experiments with *a priori* knowledge



Effective connectivity

- Definition: Causal (directed) influences between neurons or neuronal populations
- Aim of indirectly inferring on regional effects of inter-regional connectivity



Objectives for effective connectivity

Disadvantages from functional connectivity

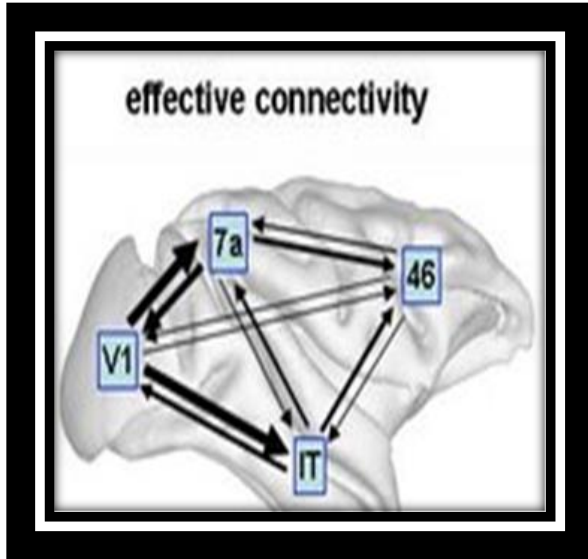
- Interpretation of findings is based on correlations; descriptive
- No mechanistic insight
- Typically, dissatisfying for experimental tasks
- Typically, suboptimal for experiments with *a priori* knowledge



Advantages to effective connectivity

- To analyse neural function on a network level
- To infer neurobiological mechanisms underlying cognitive function
- To infer causal or mechanistic function
- To infer experimental modulation
- To bridge the gap between rodent work and human work

Effective connectivity approaches

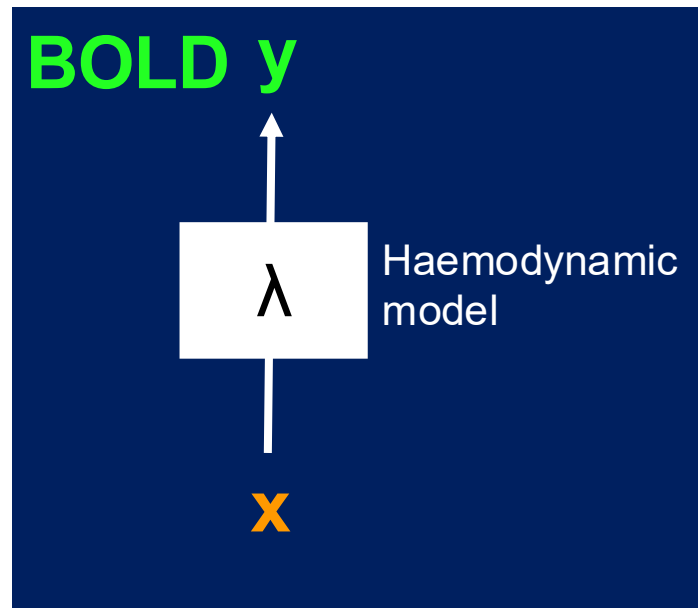


- Structural Equation Modelling
 - McIntosh et al, 1994; Büchel and Friston, 1997
- Regression models (Psychophysiological models)
 - Friston et al, 1997
- Volterra kernels
 - Friston et al, 2000
 - Time series models (Granger causality) Goebel et al, 2003
- Dynamic Causal Modelling (DCM)
 - Stephan and Friston, 2011

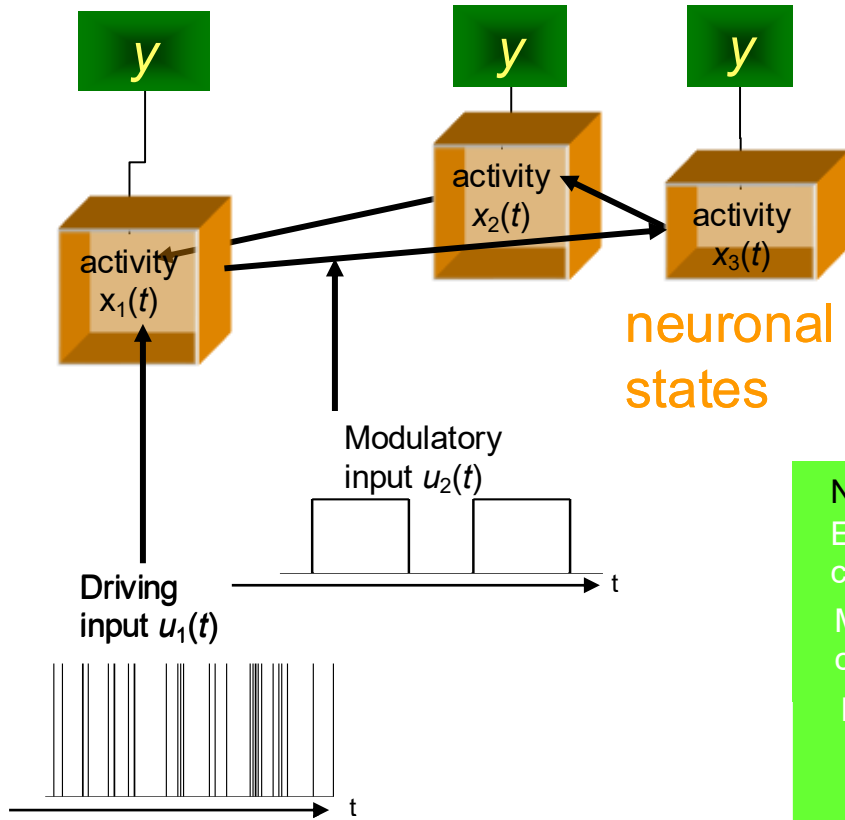
Dynamic Causal Modelling (DCM) for fMRI

Basic Idea of DCM for fMRI

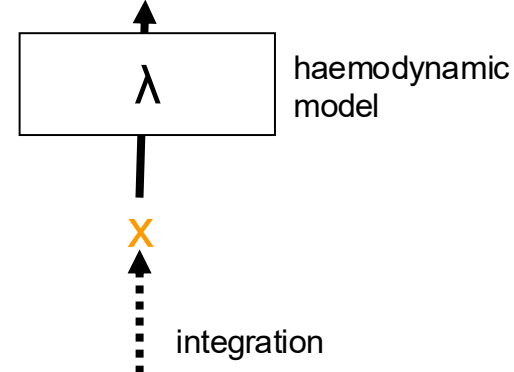
- Cognitive function is measured and modelled at the neural level.
- The modelled neural dynamics ($\mathbf{x}$) are transformed into area-specific BOLD signals ($\mathbf{y}$) by a haemodynamic forward model (λ).
- Aim: To estimate the parameters of a neural model, such that the predicted/modelled BOLD responses correspond as closely as possible to the observed/measured BOLD responses



Activity-dependent changes of neural function



BOLD y



Neural state equation

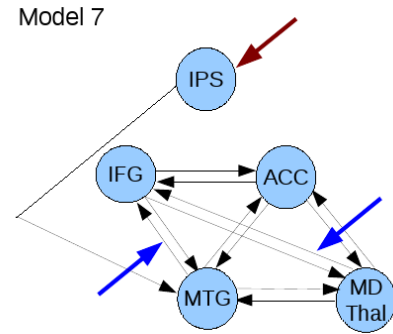
Endogenous connectivity

Modulation of connectivity

Direct inputs

No Image

No Image



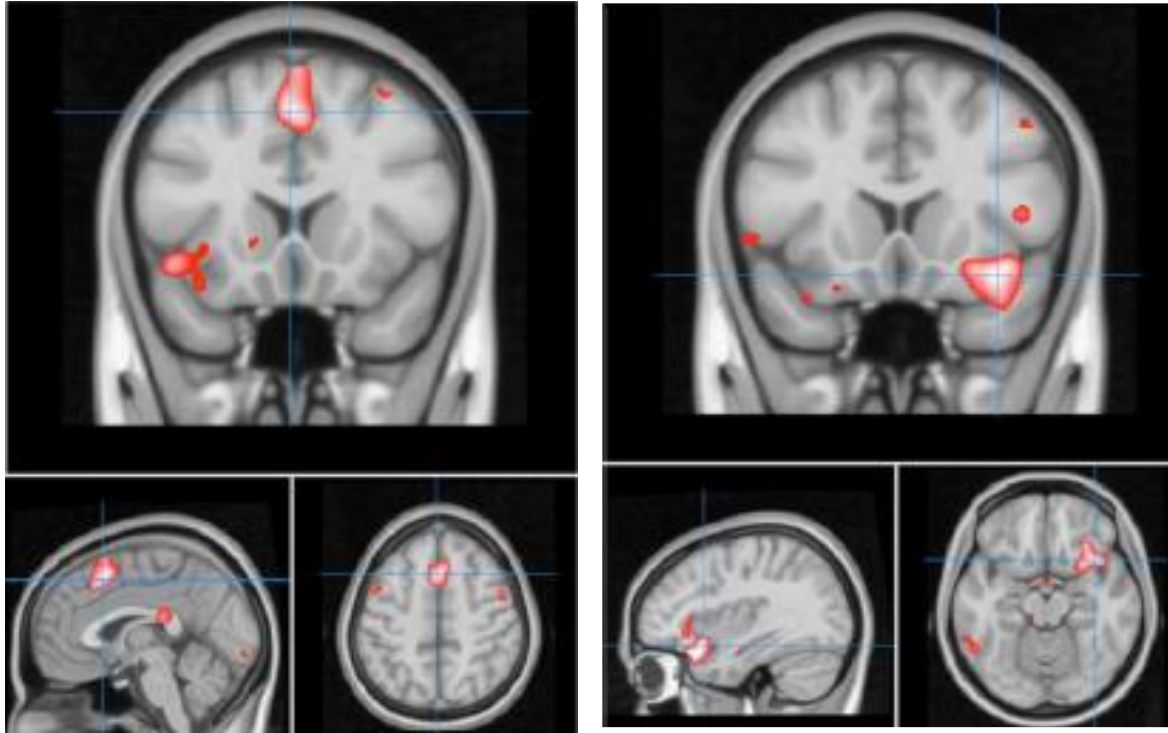
Edinburgh High Risk Study (EHRS)



- Familial risk of schizophrenia
 - At least two first-degree relatives with schizophrenia
- Healthy controls
 - No family history of schizophrenia or personal history of a serious psychiatric disorder
- Age between 16 and 25 years

HC	Healthy controls
HR	High risk subjects
HR-	High risk subjects without symptoms
HR+	High risk subjects with symptoms
4 ill	Four ill subjects with schizophrenia

Functional Connectivity



Left:
Reduced lateral-medial prefrontal
FC in HR in contrast to HC

Right:
Reduced prefrontal-thalamic FC in
HR in contrast to HC

HC Healthy controls

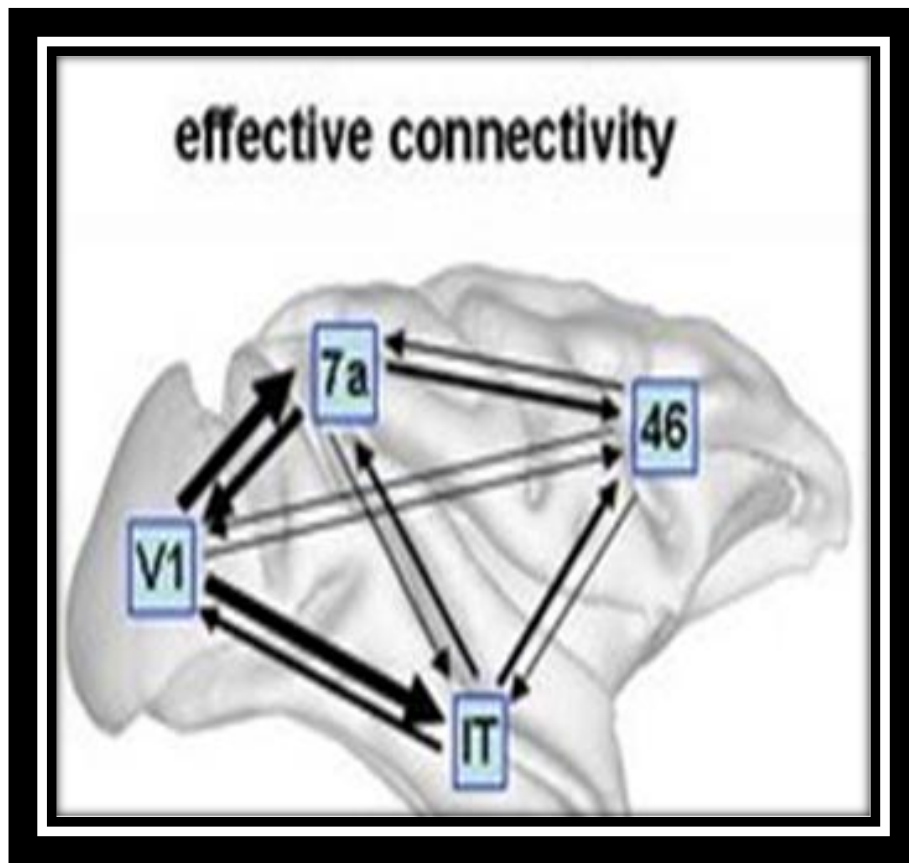
HR High risk subjects

HR- High risk subjects without
symptoms

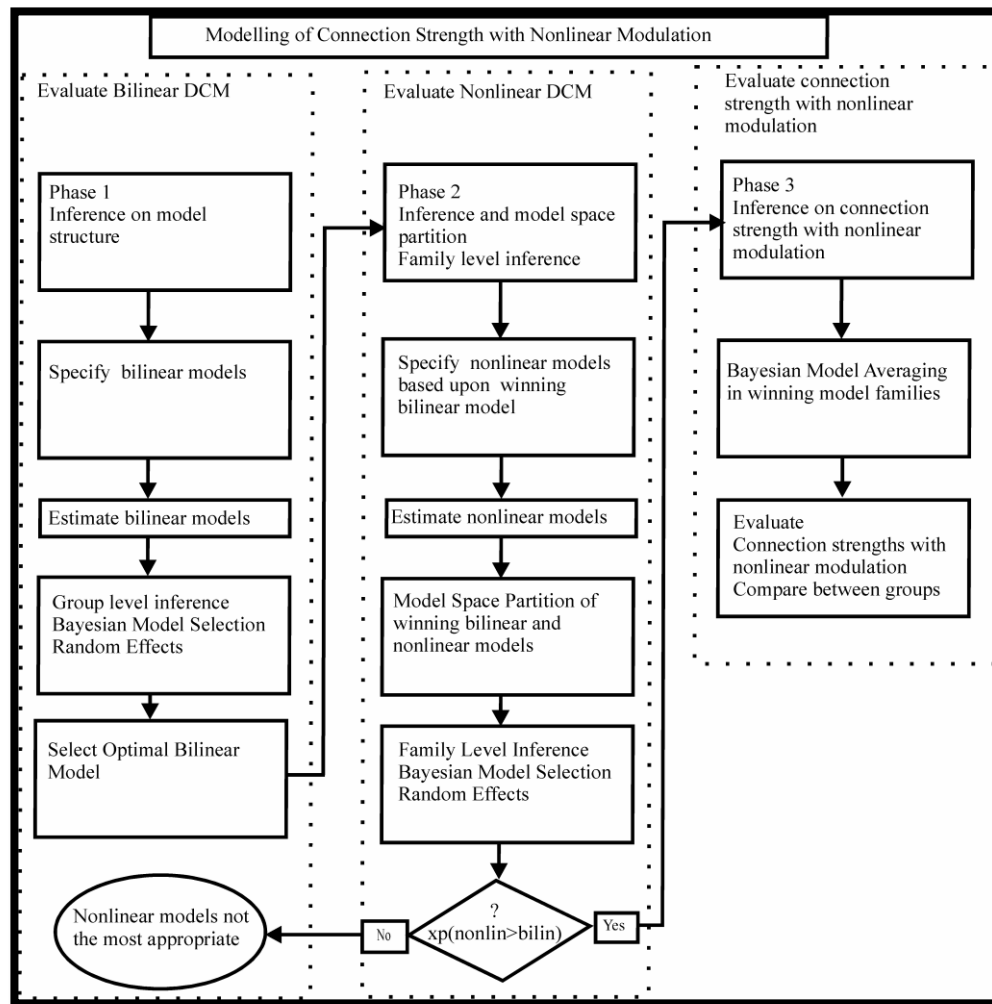
HR+ High risk subjects with symptoms

4 ill Four ill subjects with schizophrenia

Maps thresholded at $T \geq 2.5$, $P < 0.05$ corrected for multiple comparison,
minimum cluster size 50 voxels.



Bayesian estimations of how the rate of change of neuronal activity in one region influences neuronal activity in other regions

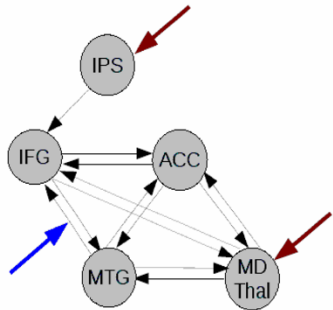


— Endogenous connection
 — Modulatory input
 — Driving input

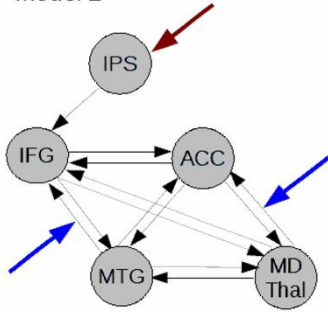
Linear Models

IPS Intra-parietal sulcus
 IFG Inferior frontal gyrus
 ACC Anterior cingulate cortex
 MTG Middle temporal gyrus
 MD Mediodorsal thalamus

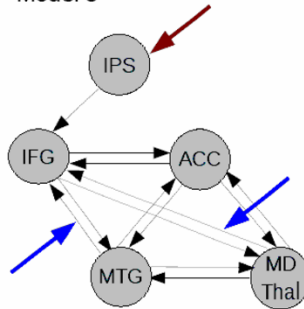
Model 1



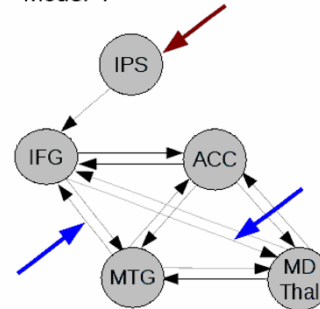
Model 2



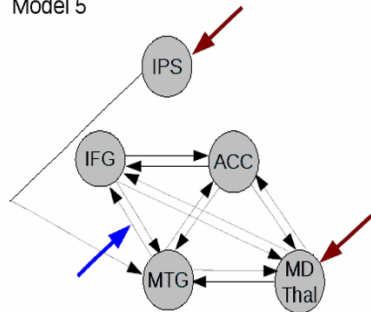
Model 3



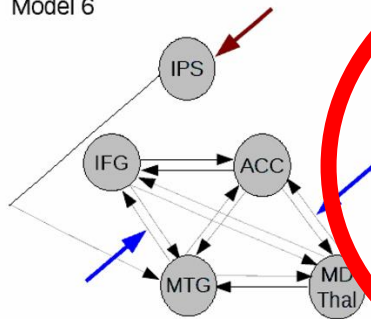
Model 4



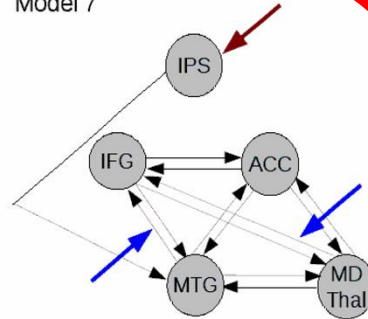
Model 5



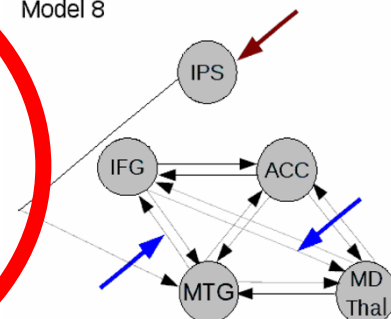
Model 6



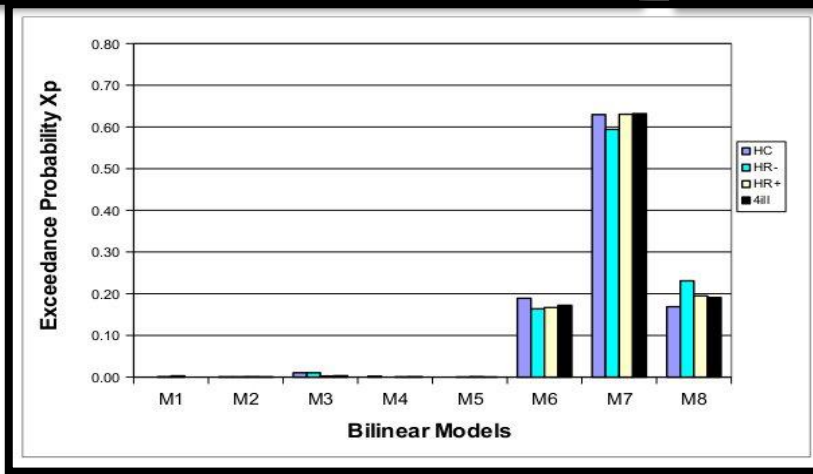
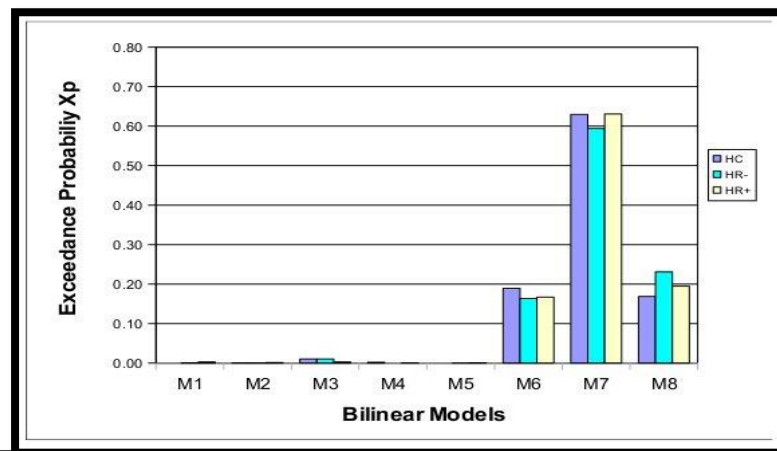
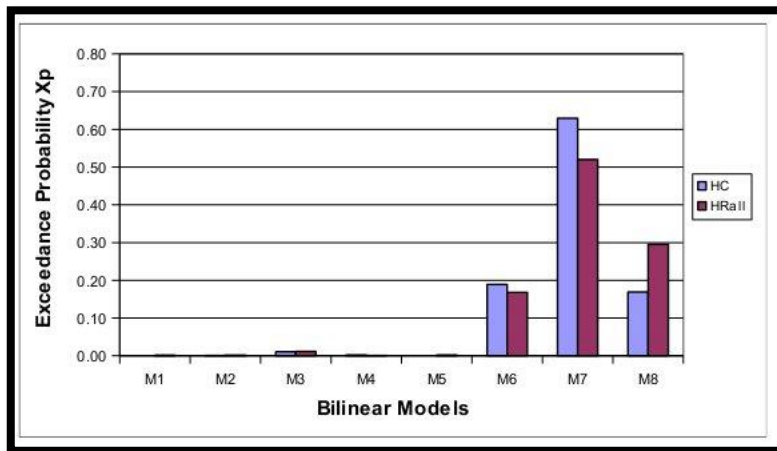
Model 7



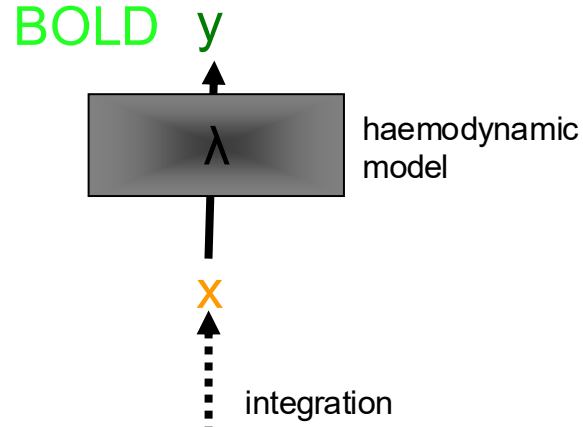
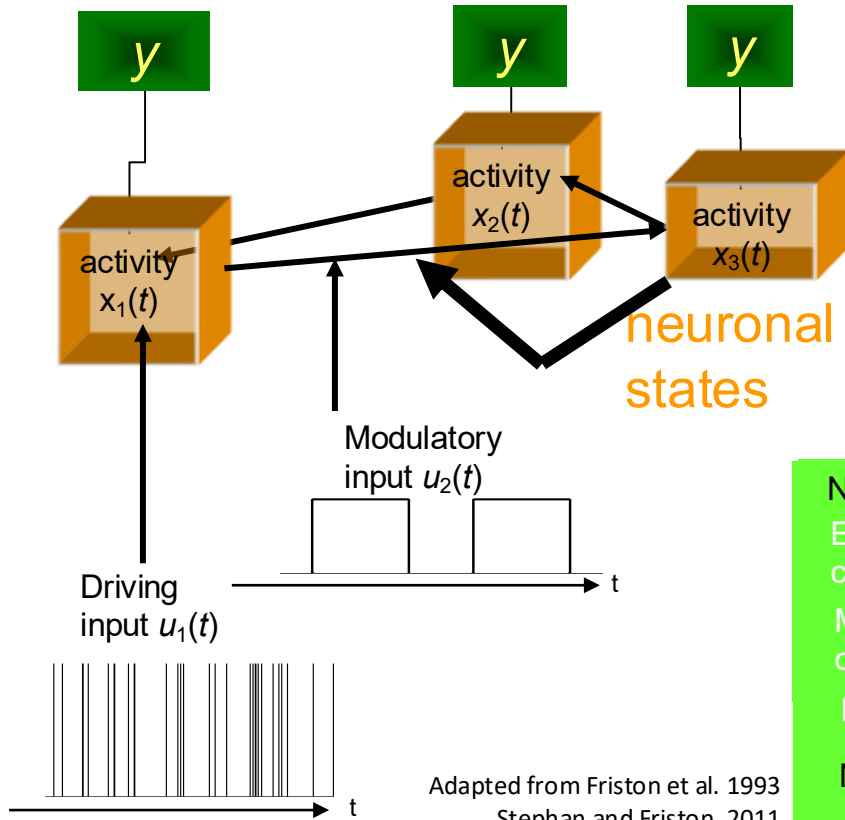
Model 8



Optimal linear model



Activity-dependent changes of neural function



Neural state equation

Endogenous connectivity

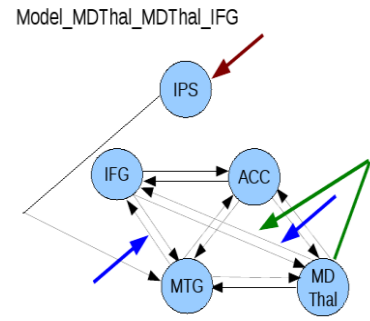
Modulation of connectivity

Direct inputs

Nonlinear input

$$D^{(j)} = \left. \frac{\partial^2 f}{\partial x_j^2} \right|_{u=0} \quad (1 \leq j \leq n)$$

Adapted from Friston et al. 1993
Stephan and Friston, 2011

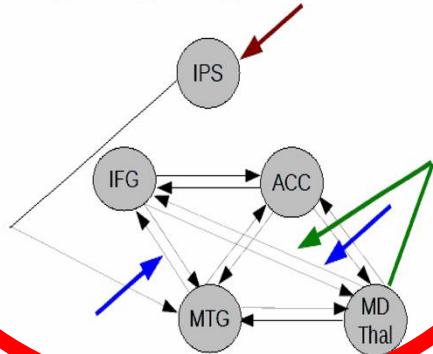


Nonlinear models

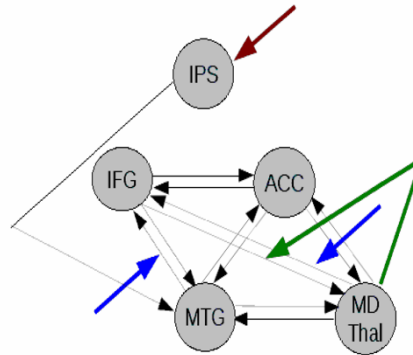
IPS Intra-parietal sulcus
 IFG Inferior frontal gyrus
 ACC Anterior cingulate cortex
 MTG Middle temporal gyrus
 MD Thalamus Mediodorsal thalamus

— Endogenous connection
 — Modulatory input
 — Driving input
 — Nonlinear modulation

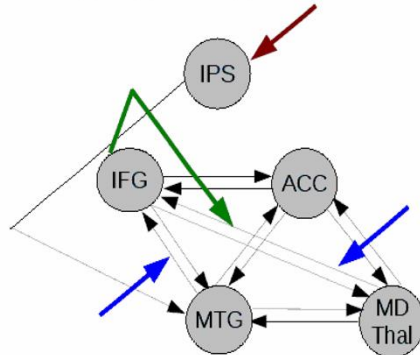
Model_MDThal_MDThal_IFG



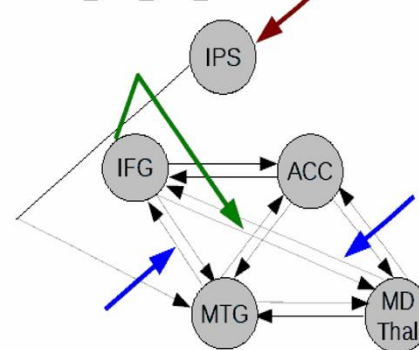
Model_MDThal_IFG_MDThal



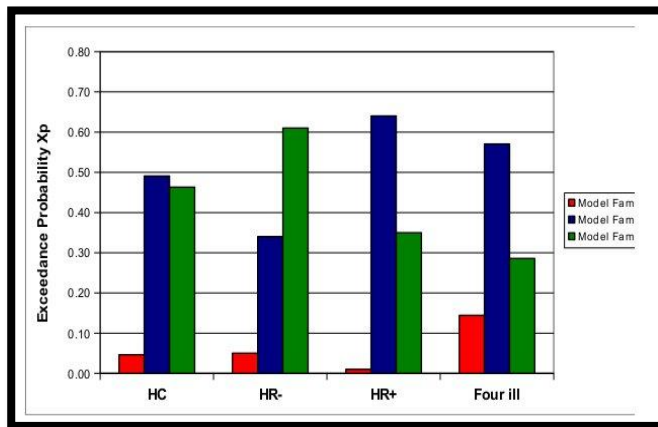
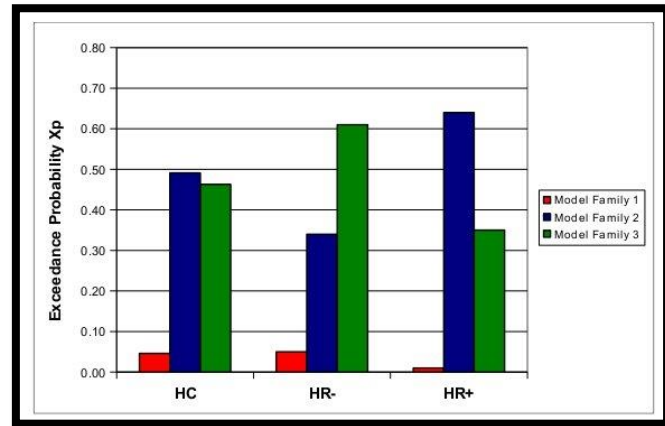
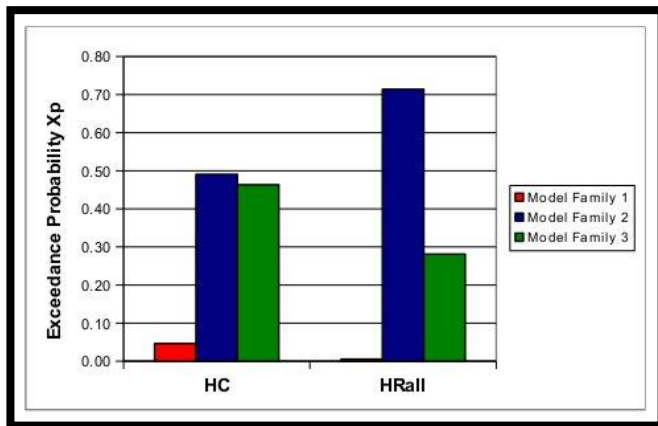
Model_IFG_MDThal_IFG



Model_IFG_IFG_MDThal

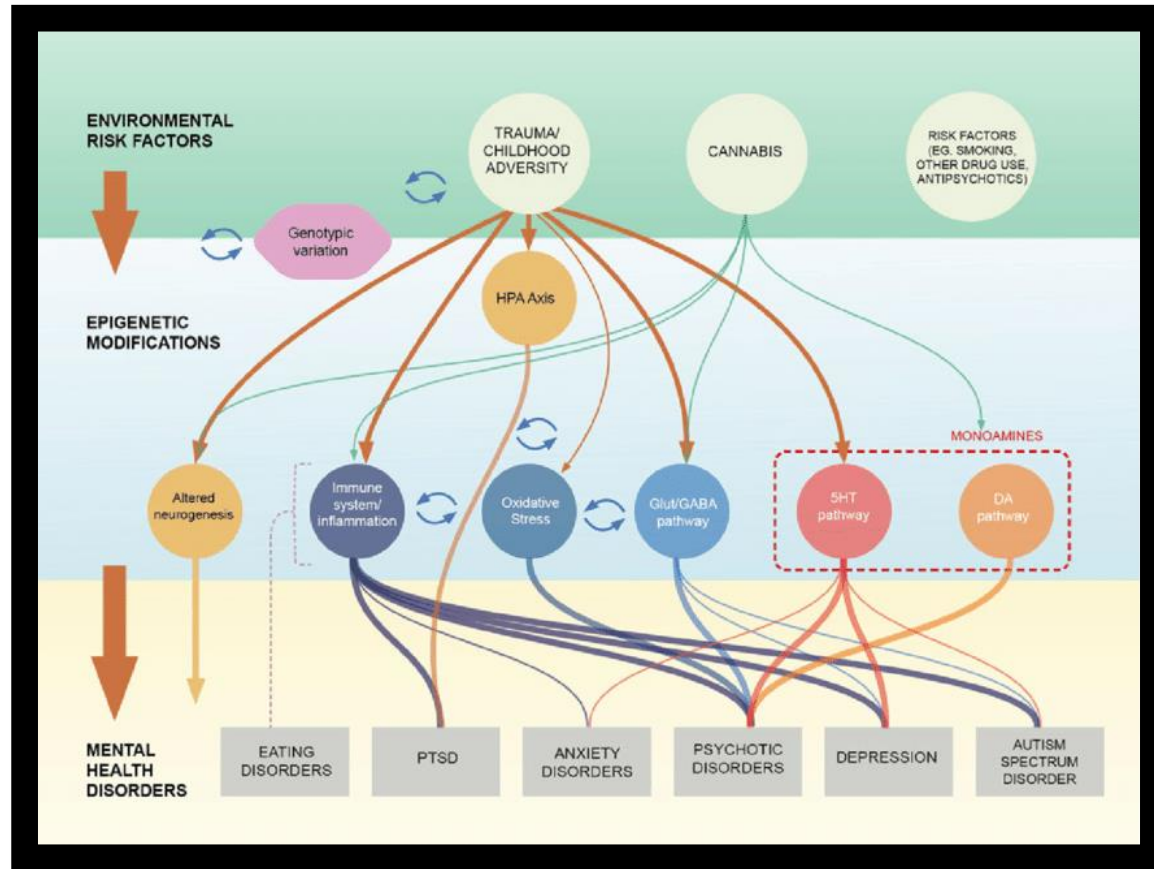


Optimal nonlinear model family



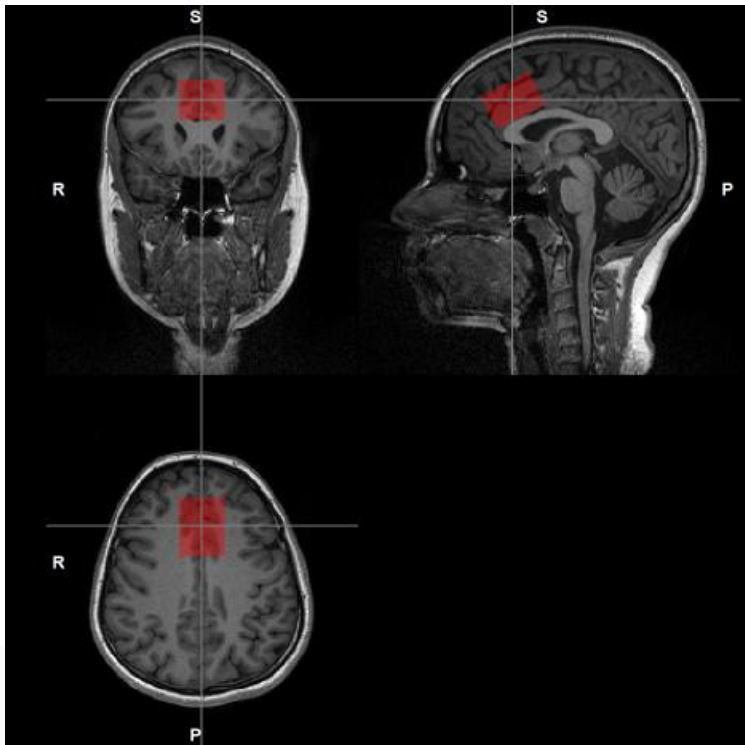
Biopsychosocial study

- Young healthy adults
 - No current mental or physical health diagnosis or treatment
 - No past history of psychotic disorders
 - No/low childhood adversity experience versus moderate/high childhood adversity experience
- Age between 18 and 32 years



Glutamate and GABA concentrations

dorsal anterior cingulate cortex (dACC)



Corcoran, ... Dauvermann, 2020;
Siegel-Ramsay, ... Dauvermann, 2020

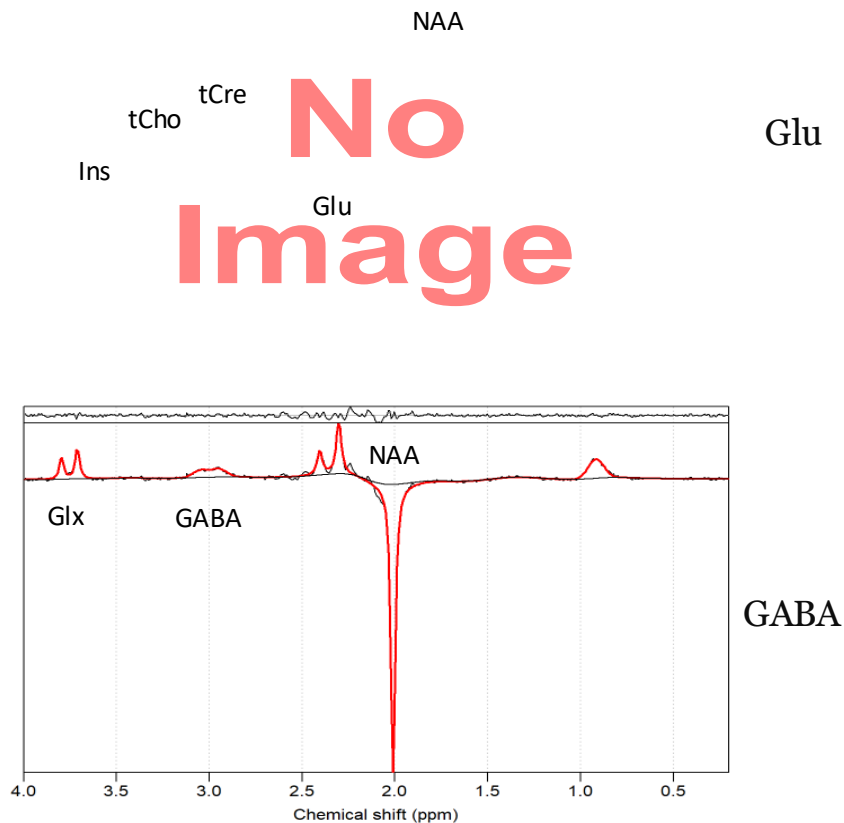
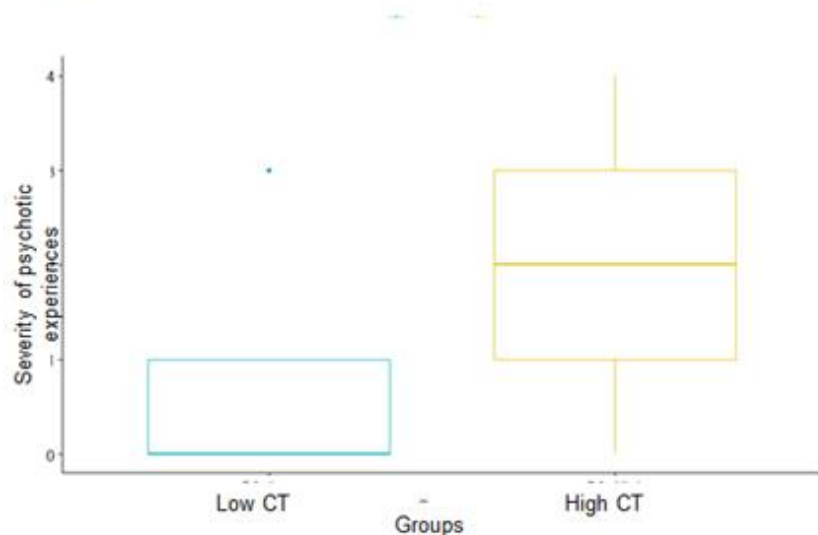
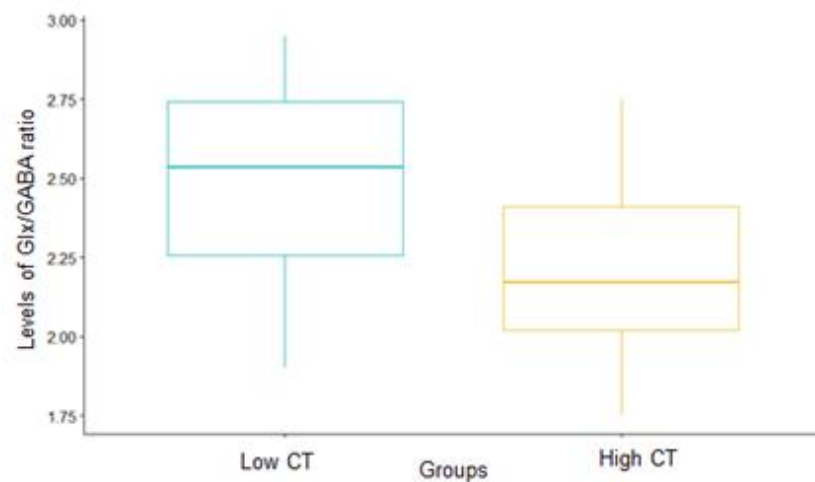


Figure 2. Childhood trauma, subthreshold clinical symptoms and Glx/GABA ratio in the dorsal anterior cingulate cortex (dACC)



A. Subthreshold clinical symptoms

Young healthy adults with high childhood trauma levels reported higher subclinical levels psychotic experiences ($T_{(28)} = 2.538$, $p < 0.017$) when compared to individuals with low levels.

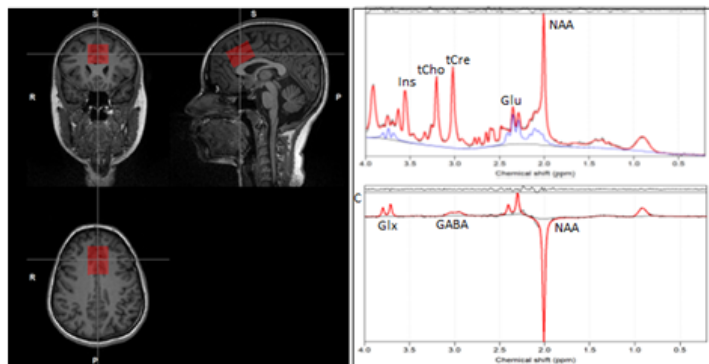


B. Childhood trauma and Glx/GABA ratio in the dACC

Young healthy adults with high childhood trauma levels showed lower levels of Glx/GABA ratio in the dorsal anterior cingulate cortex (dACC) compared to individuals with low levels of childhood trauma ($T_{(28)} = 2.883$, $p < 0.007$).

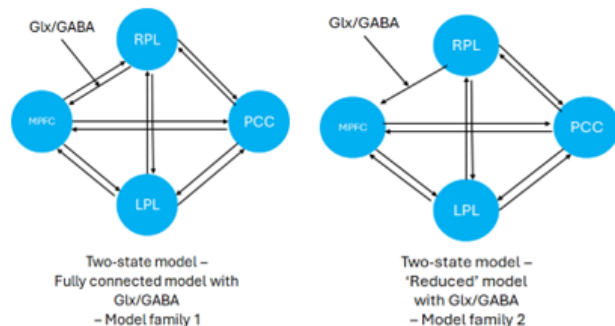
C Development and validation of model of excitatory-inhibitory balance measure

C1 Excitatory and inhibitory neurotransmitters concentrations



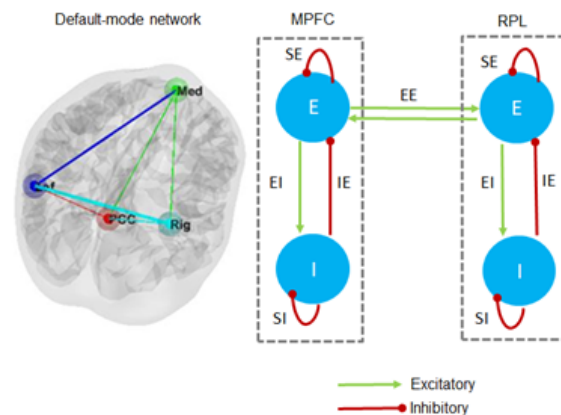
- **Component:** Glutamate and GABA concentrations in dACC
- **Measure:** Glutamate/GABA ratio between pre and post acute stress
- **Modality:** MRS; sLASER for Glutamate; MEGA-PRESS for GABA

C3 Combination of both excitatory and inhibitory neurotransmitters concentrations and connection strengths



- **Component:** Glutamate and GABA concentrations in dACC
- **Measure:** Δ Glutamate+Glutamine/GABA ratio between pre and post acute stress
- **Modality:** MRS; sLASER for Glutamate and Glutamate+Glutamine; MEGA-PRESS for GABA

C2 Excitatory and inhibitory connection strengths - Dynamic Causal Modelling for fMRI

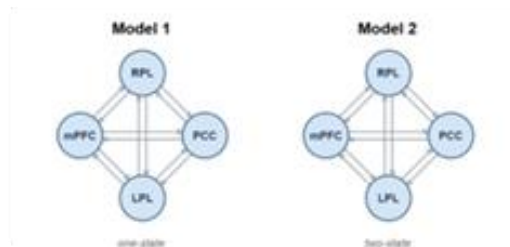


- **Component:** Excitatory and inhibitory connection strengths
- **Measure:** Inferred excitatory and intrinsic inhibitory connection strengths
- **Modality:** resting-state fMRI

Model space

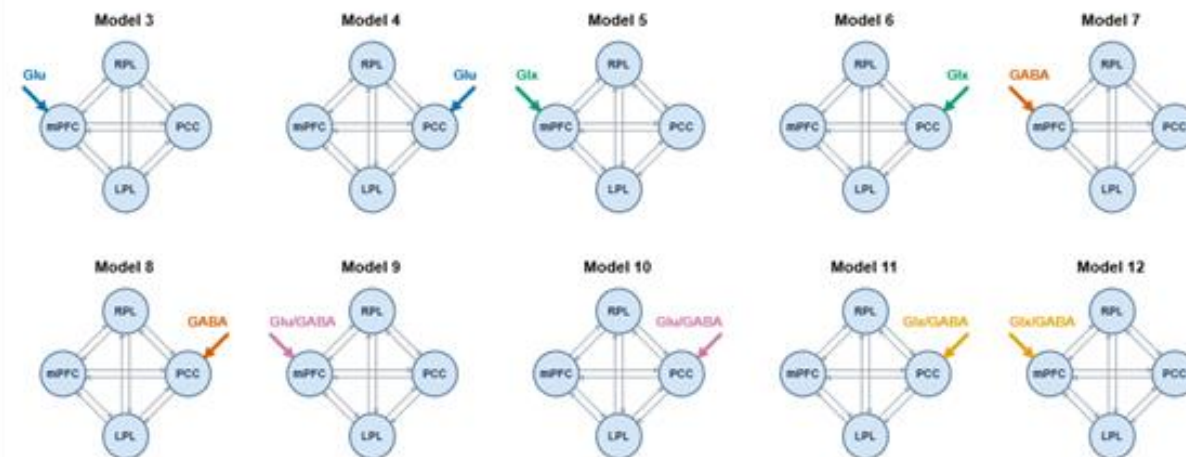
mPFC medial PFC
RPL right parietal lobe
LPL left parietal lobe
PCC posterior cingulate cortex

A

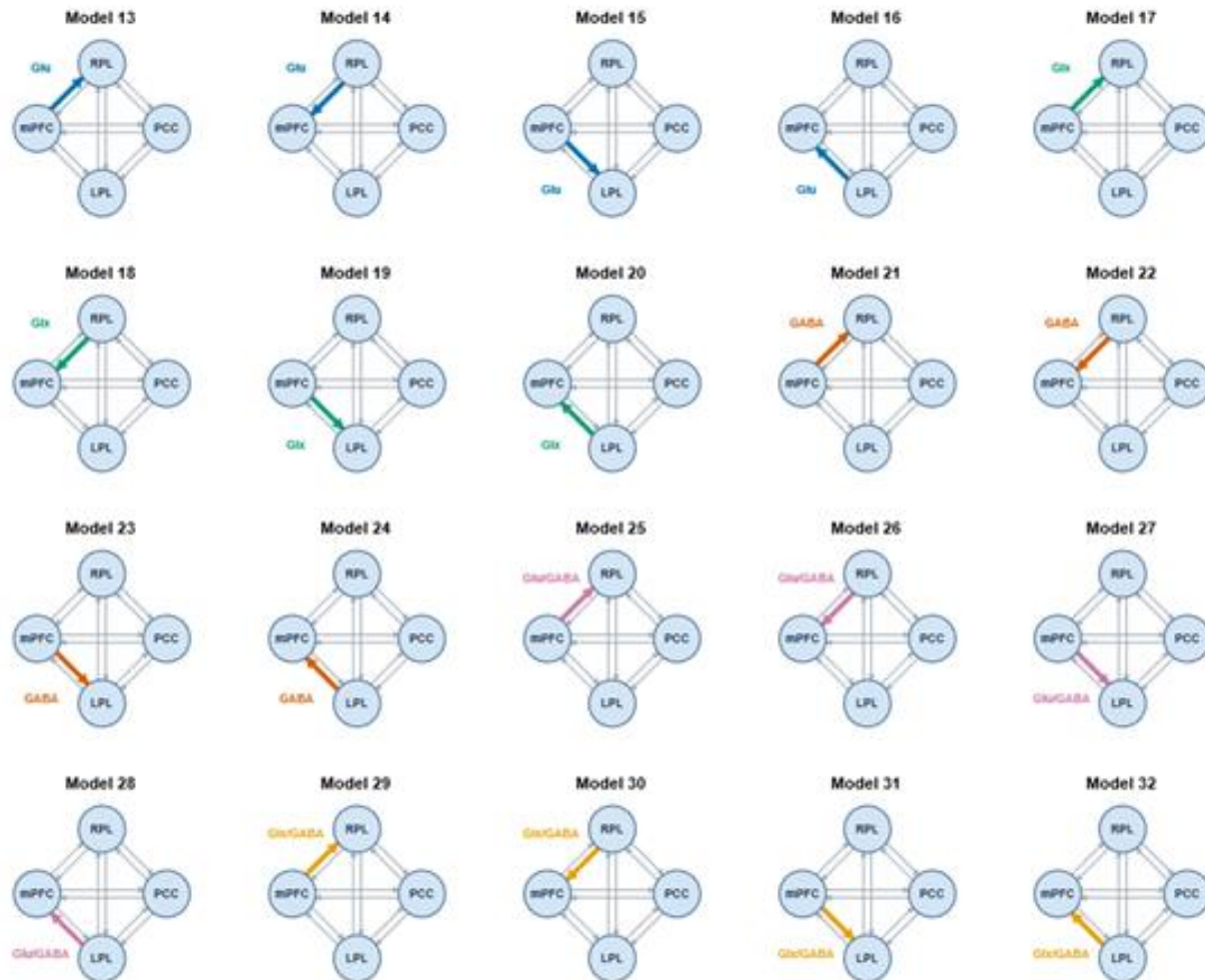


— Endogenous connection
— Driving input

B



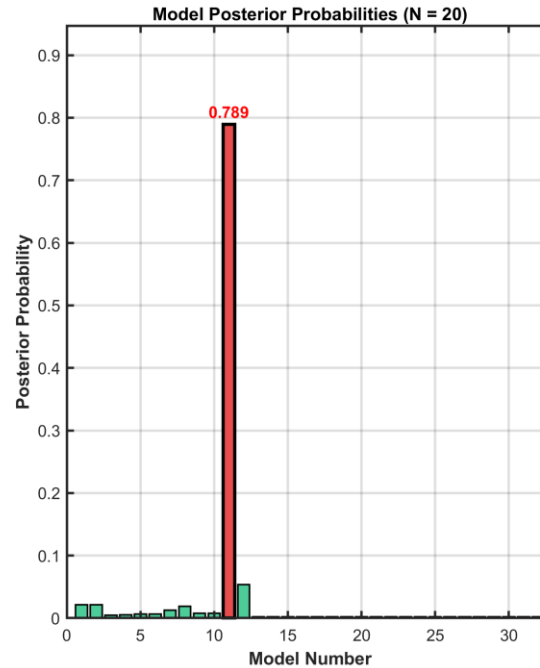
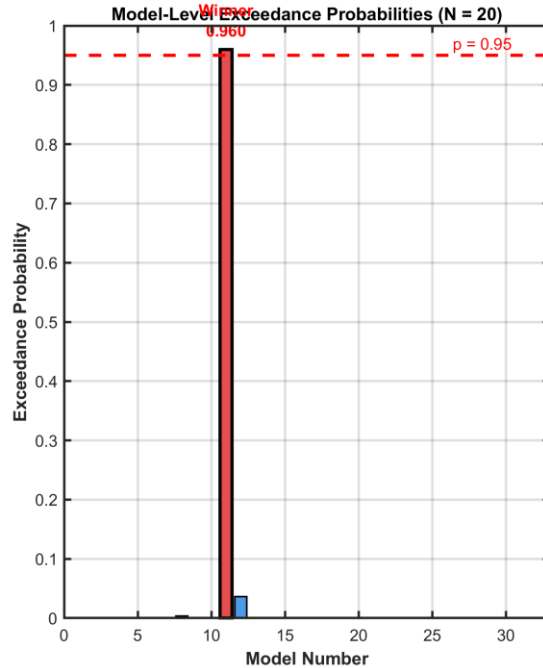
C



mPFC medial PFC
 RPL right parietal lobe
 LPL left parietal lobe
 PCC posterior cingulate cortex

— Endogenous connection
 — Modulatory input

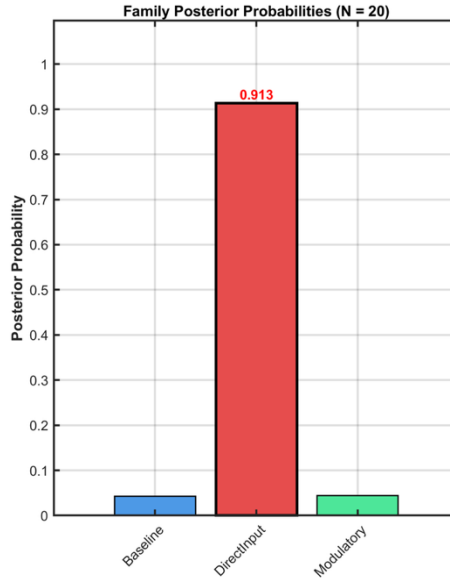
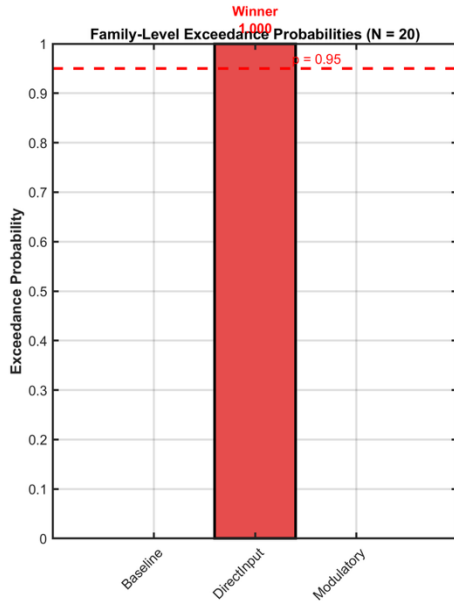
Model-level random-effects Bayesian Model Selection across the 32-model space (N = 20)



Left: exceedance probabilities for each candidate model; Model 11 (red bar) was the only model to exceed the 0.95 threshold (dashed red line), with an exceedance probability of 0.960.

Right: posterior probabilities for each model; Model 11 again accounted for the greatest share of the evidence (0.789).

Family-level random-effects Bayesian Model Selection (N = 20)



Left: exceedance probabilities for the Baseline (Models 1–2), DirectInput (Models 3–12), and Modulatory (Models 13–32) families; the DirectInput family (red bar) achieved a perfect exceedance probability of 1.000, well above the 0.95 threshold (dashed red line).

Right: posterior probabilities for each family, confirming the dominance of the DirectInput family (0.913).

Discussion





Can the application of functional and/or effective connectivity to brain imaging contribute to clinical and cognitive diagnostic, predictive and treatment interventions in the future?



Summary

- Main principle of functional connectivity:
 - Analyse data between two brain regions with temporal correlations
- Main principle of DCM for effective connectivity:
 - Analyse data and generative models in a Bayesian framework to infer parameters and compare models
- Advantages and disadvantages dependent on available data, hypothesis and *a priori* findings in the literature

Thank you very much!

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